INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Profit, Loss, Discount & Alligations

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Profit, Loss, Discount & Alligations

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/aptitude/profit-loss/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 01_Mathematical_Ability_Technical_Aptitude
Total Verified Questions Extracted: 10
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4, quiz-5, quiz-6, quiz-7

📋 Section 1: Rules to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- Profit = SP - CP; Loss = CP - SP.
- Profit % = (Profit / CP) * 100; Loss % = (Loss / CP) * 100 (Always calculated on Cost Price).
- SP = CP * (100 + P%) / 100; CP = [100 / (100 + P%)] * SP.
- Discount is always calculated on Marked Price (MP): Discount = MP - SP; Discount % = (Discount / MP) * 100.
- SP = MP * (100 - D%) / 100.

⚡ Section 2: How to Solve Quickly

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🚀 Section 3: Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- Relation between MP and CP: (MP / CP) = (100 + P%) / (100 - D%).

- When two articles are sold at the same selling price, one at a gain of x% and the other at a loss of x%, there is always an overall loss of (x/10)^2 %.
- Dishonest dealer using false weight: Gain % = [Error / (True Value - Error)] * 100%.

📚 Section 4: Practice Questions, Options & Step-by-Step Solutions

Contains **10** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
If the list price of a book is reduced by Rs. 5, then a person can buy 5 more books for Rs. 300. The original cost of the book is

Options:
- **(A)** `Rs. 15`
- **(B)** `Rs. 20`
- **(C)** `Rs. 25`
- **(D)** `Rs. 30`

Correct Answer: (B) `Rs. 20`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
If x is earlier price, then


300/(x-5)- 300/x = 5


solving this , we get, x=20 or -15 out of which only 20 is acceptable.
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:
A shopkeeper sells 18 mangoes for the purchase price of 20 mangoes. The percent profit made by the shopkeeper is.

Options:
- **(A)** `10%`
- **(B)** `11.11%`
- **(C)** `9.09%`
- **(D)** `12%`

Correct Answer: (B) `11.11%`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
let c.p of 1 mango Re 1.

then c.p of 18 mangoes= Rs 18

c.p of 20 mangoes Rs 20 = s.p of 18 mangoes

profit= s.p of 18 mangoes - c.p of 18 mangoes

=20 - 18

= 2

profit %= 2/18*100

=11.11 ans
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:

A merchant marks his goods up by 75% above his cost price. What is the maximum % Amount that he can offer so that he ends up selling at no profit or loss?

Options:

- **(A)** `75%`
- **(B)** `46.67%`
- **(C)** `300%`
- **(D)** `42.85%`

Correct Answer: (D) `42.85%`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let’s say the C.P. of the goods is 100.


So marked price = 1.75 * 100 = 175


If the merchant doesn’t want any profit or loss, he may want to see the goods at Rs 100 only. i.e. he may want to give discount of Rs 75 at marked price of Rs 175.


Let’s say it’s x% discount


so 75 = x% of 175


75 = x * 175 / 100


x = 75 * 100/175 = 300/7 = 42.857 %


 
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:
Profit and loss problem: I bought a book for Rs 60 , I sold off to a friend for Rs 70 but after a while i felt sorry , that i sold it and bought it back for Rs 80, how much loss did I incur?

Options:
- **(A)** `12`
- **(B)** `30`
- **(C)** `20`
- **(D)** `10`

Correct Answer: (D) `10`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
cp of book=60


sp of book=70


profit=10


again cp=80


then loss =80-70=10
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:
Lucia is a wonderful grandmother.Her age is b/w 50 to 70. Each of her sons have as many sons as they have brothers. Their combined number gives Lucia's present age?what is her age?

Options:
- **(A)** `a)85`
- **(B)** `b)55`
- **(C)** `c)84`
- **(D)** `d)64`

Correct Answer: (D) `d)64`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
If she has x sons


each son has x-1 brothers.


Total no of grandchildren=x(x-1)


Total No OF Sons+Grandchildren= x+x(x-1)=x^2


=&gt;Lucia\'s age is a perfect square between 50 and 70.


The only number that satisfies this condition is 64.
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:
If a 10 lit. mixture contains milk and water in the ratio 2:1 then, how much more mixture should be added to change the ratio to 1:2 ?

Options:
- **(A)** `8`
- **(B)** `9`
- **(C)** `10`
- **(D)** `11`

Correct Answer: (C) `10`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
If you add the same mixture again,the ratio will still remain the same.

The question should be how much water to be added

In 10 litres, let quantity of milk = 2x and quantity of water = x

=&gt; 2x + x = 10

=&gt; x = 10/3 , 2x = 20/3

Let y litres of water be added to make the ratio 1:2

=&gt; (20/3) / (10/3 + y) = 1 / 2

=&gt; 40/3 = 10/3 + y

=&gt; y = 10 liters

Therefore,10 liters of water to be added.
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:
Suresh invested a sum of Rs. 15000 at 9 percent per annum Simple interest and Rs. 12000 at 8 percent per annum compound interest for a period of 2 years. What amount of interest did Suresh earn in 2 years?

Options:
- **(A)** `5120`
- **(B)** `3574`
- **(C)** `4893`
- **(D)** `4696.8`

Correct Answer: (D) `4696.8`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
S.I = PTR/100

S.I =15000*9*2/100

S.I= 2700

C.I =P(1+r/100)^n -P

C.I= 12000(1+8/100)^2-12000

C.I=1996.8

total amount in 2 years

as interest = S.I +C.I

= 2700+1996.8

=4696.8
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:
Tea worth Rs. 126 per kg and Rs. 135 per kg are mixed with a third variety in the ratio 1 : 1 : 2. If the mixture is worth Rs. 153 per kg, the price of the third variety per kg will be:

Options:
- **(A)** `Rs. 169.50`
- **(B)** `Rs. 170`
- **(C)** `Rs. 175.50`
- **(D)** `Rs. 180`

Correct Answer: (C) `Rs. 175.50`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Then (126*1+135*1+2*x)/(1+1+2)= 153

on solving

261 + 2x = 4x153

2x = 612-261

x=351/2

X= 175.50
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:
A milk vendor has 2 cans of milk. The first contains 25% water and the rest milk. The second contains 50% water. How much milk should he mix from each of the containers so as to get 12 litres of milk such that the ratio of water to milk is 3 : 5?

Options:
- **(A)** `4 litres, 8 litres`
- **(B)** `6 litres, 6 litres`
- **(C)** `5 litres, 7 litres`
- **(D)** `7 litres, 5 litres`

Correct Answer: (B) `6 litres, 6 litres`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let the cost of 1 litre milk be Rs 1.

milk in 1 litre mixture in 1st can = 3/4 litre

C.P.  of  1 litre  mixture  in 1st can = Rs 3/4 milk

in 1 litre mixture in 2nd can = 1/2 litre  milk

in 1 litre mixture in 2nd can = Rs 1/2 milk

in 1 litre of final mixture = 5/8 litre

Mean price = Rs 5/8 by the rule of alligation we have

x/y = ( 3/4 - 5/8)/(5/8 - 1/2)

x:y = 1/8: 1/8

x/y = 1/1

therefore we will mix 6 litre from each can
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:
In what ratio must a grocer mix two varieties of tea worth Rs. 60 a kg and Rs. 65 a kg so that by selling the mixture at Rs. 68.20 a kg he may gain 10%?

Options:
- **(A)** `3 : 2`
- **(B)** `3 : 4`
- **(C)** `3 : 5`
- **(D)** `4 : 5`

Correct Answer: (A) `3 : 2`

PrepInsta Detailed Explanation & Mathematical Derivation:

S.P. of 1 kg of the mixture = Rs. 68.20, Gain = 10%.

C.P. of 1 kg of the mixture = Rs. (100/110*68.20) =Rs. 62

By the rule of alligation, we have:

Cost of 1 kg tea of 1st kind. = Rs. 60,

Cost of 1 kg tea of 2nd kind.

= Rs. 65,

Mean Price = Rs. 63

i.e., the difference is of Rs. 3 and Rs. 2.

therefore, the ratio is 3:2